

A Ship Target Detection Method for SAR Image Based on Local Region

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Abstract—Using the knowledge of image processing, machine learning and pattern recognition to develop the techniques of acquiring target regions from Synthetic Aperture Radar (SAR) images accurately has caught researchers' wide attention. A ship target detection method for SAR image based on local region under the multi-scale and multi-target condition is introduced in this paper. This method firstly learns the gradient-based local region generation mode based on the training data with object annotation, then generates a small number of local regions with different sizes, and finally uses the local region based CFAR detector, where the extracted object regions are regarded as the guard windows instead of setting fixed guard window to detect the true object regions. Due to the introduction of local regions, the proposed method can obtain good detection performance in the multi-scale situation, and directly obtain the accurate target regions to avoid the problem caused by the target clustering in traditional ship detection method. The effectiveness of the proposed method is verified using measured RADASAT-2 data by comparing with the traditional SAR ship target detection methods.

Keywords—local regions, constant false alarm rate (CFAR), synthetic aperture radar (SAR) images, target detection.

I. INTRODUCTION

Target detection for synthetic aperture radar (SAR) images is a subtask in SAR automatic target recognition (ATR). A typical SAR ATR system often consists of three stages: detection, discrimination, and classification or recognition [1]. As the first step of the whole SAR ATR system, target detection has great influence on the successive processing[2]. Synthetic aperture radar (SAR) image is an active radar which can provide high-resolution images in the microwave band under all weather conditions. Since SAR images are widely used in ship traffic monitoring, fishing vessel detection, and immigration control[3][4], ship detection has been an important application of SAR image. Ship detection in SAR image has been studied extensively, and numerous studies have been performed to develop ship detection algorithms in SAR images[3][5][6][7], of which the two-parameter constant false alarm rate (CFAR) method is often used for ship target detection in SAR image[3][6][7].

The Two-parameter CFAR is actually a cell averaging CFAR (CA-CFAR) technique based on the assumption of Gaussian background[8]. Two-parameter CFAR needs to know the size of the detected ship first, only after which the size of the detecting windows can be determined. It works well in situations where a single target is present in locally homogeneous clutter. However, in multi-scale and multi-target situations, the fixed guard window and background window will contain one or more interfering target pixels,

which can result in degradation of the its detection performance. Moreover, it uses a sliding window to detect the possible existing ship pixels and needs much time for the statistical computation of clutter background[2][3][5]. So the pixel-based method is more computationally expensive. In addition, the detected pixels must be remerged into individual ship target finally, which is a difficult task when ships are close. it will result two or more target pixels remerged into a target in multi-scale and multi-target situations, which will reduce the detection performance and bring much difficulties to following discrimination and recognition. To improve the detection performance of Two-parameter CFAR for ship target detection in multi-scale and multi-target situations and accelerate its detection speed, the local regions are introduced.

In recent years, local region generation [9][10][11] has become a promising technique for many vision recognition tasks, especially for class-specific object detection in color images. Instead of handling with tremendous amount of bounding boxes in sliding windows fashion, local region generation selects much fewer candidate bounding boxes (typically from hundreds to a few thousands per image) that cover most of the objects in the image. This candidate bounding boxes are called local regions. This technique can speed up the computation by reducing the candidate bounding boxes and improving the detection accuracy by allowing the usage of more sophisticated learning machinery[12][13].

To make full use of local region generation technique, based on local regions we propose a ship target detection algorithm based on local regions for SAR image in this paper. We firstly use the local region generator to generate a small set of local regions rapidly. Then we regard local region as guard window, rather than setting a fixed guard window, and deal with the local region as a whole and use the two-parameter CFAR method to detect it. Finally, the detection result is given by non-maxima suppression[14].

The rest of this letter is organized as follows. Section II describes the proposed ship target detection approach. Section III provides some experimental results of the proposed approach and Section IV concludes this letter.

II. PROPOSED METHOD

A. Local region Generation Technique

In recent years, local region generation has become a promising technique for many vision recognition tasks, the binarized normed gradients (BING) [15] algorithm is one simple and efficient method among the methods generating

local regions. This method is based on the observation that generic objects with well defined closed boundary can be discriminated by looking at the norm of gradients, with a suitable resizing of their corresponding image windows into a small fixed size. Based on the observation, it resizes a image window to 8×8 and use the norm of the gradients (NG) as a simple 64D feature for learning a generic objectness measure in a two stages cascaded SVM framework. In order to accelerate the speed of the local regions generating, it resizes a image window to 8×8 and uses the norm of the gradients (NG) as a simple 64D feature to learn a generic objectness measure in a two stages cascaded support vector machine (SVM) framework.

When resizing ship target windows to the fixed size 8×8 , the simple norm of the gradients in intensity space becomes a good discriminate feature. As shown in Fig. 1, though the ships have different scales and aspect ratios, they do share strong correlation in normed gradient space. The normd gradients g is calculated in (1)

$$g = \min(\|g_x\| + \|g_y\|, 255) \quad (1)$$

where the image horizontal gradients g_x and vertical gradients g_y are calculated via the convolution of the 1-D mask $A = [-1, 0, 1]$ and A^T with the image in intensity space.

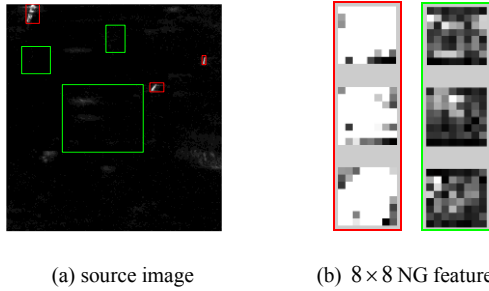


Fig. 1. object (red) and non-object (green) windows and their corresponding normed gradients, i.e. a 8×8 NG feature

The NG feature and its binarized normed gradients BING feature can be used to discriminate the object from non-object, which can be utilized to predict the existence of objects effectively.

As shown in Fig. 2, to generate a small set of bounding boxes as our local regions. Firstly we need to learn an objectness measure mode $\mathbf{W} \in \mathbb{R}^{64}$ using linear SVM with training images. We select the labeled ground truth bounding boxes and random sample bounding boxes with more than 0.5 intersection-over-union (IoU) overlap with a ground-truth box as positive training samples. Random sampled background bounding boxes with less than 0.2 IoU overlap with a ground-truth box are used as negative training samples. Each training sample is resized to fixed size 8×8 and represented by the NG feature. Then, we resized the image to a different quantized sizes (scales and aspect ratios) which are predefined and then each resized image is scored with a linear mode $\mathbf{W} \in \mathbb{R}^{64}$,

$$s_i = \langle \mathbf{W}, \mathbf{g}_i \rangle \quad (2)$$

$$l = (i, x, y) \quad (3)$$

where s_i , $\mathbf{g}_i$, l , i , and (x, y) are filter score, NG feature, location, size and position of a window, respectively. Finally we select a small set of local regions from each size i using non-maximal suppression (NMS).

To speed up the extraction of the local regions, we adopt the same binary approximation method used in [15] to approximate the learned linear model $\mathbf{W} \in \mathbb{R}^{64}$ and NG feature.

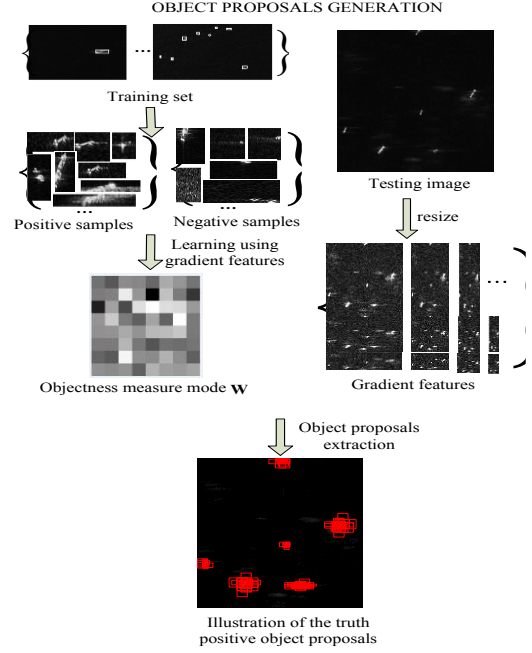


Fig. 2. The framework of the proposed target region generation

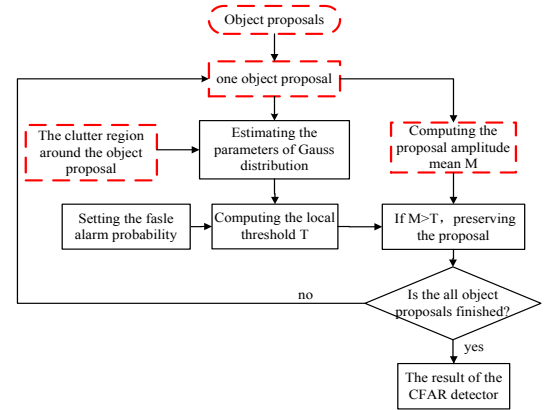


Fig. 3. Flow Chart of the target regions based CFAR algorithm

B. CFAR Detection based on Local regions

Local regions generation is only the pre-processing of the detection. The object detection rate is high, but the precision is low. In the subsequent detection, we need to remove the false-positives as well as improve the precision. The local regions around the targets are compact bounding boxes. We regard each local region as a whole and use the average pixel intensity in the bounding box to represent it. Then the two-parameter CFAR is used to determine whether the bounding

box is a target region. The local regions based CFAR detection process is shown in Fig. 4.

The two-parameter CFAR method is often used for ship targets detection in the SAR image. However, the two-parameter CFAR method needs to know the size of the detected ship first, in multi-scale and multi-target situations, fixed target window, guard window and background window will affect the detection performance. Moreover, the traditional two-parameter CFAR uses a sliding window to detect the possible existing ship pixels and need much time for the statistical computation. Different to the traditional two-parameter CFAR, we regard the local region as the target window and detect the possible local regions to deal with multi-scale and multi-target situations and reduce the time of the statistical computation. Under the assumption that sea clutter are Gaussian distributed, a local region can be preserved as candidate target region whose amplitude mean ME stand out from the surroundings according to

$$\begin{array}{c} \text{target} \\ \text{ME} > \text{Th} \\ \text{clutter} \end{array} \quad (4)$$

$$\text{Th} = \hat{\mu}_c + \hat{\sigma}_c \cdot K \quad (5)$$

$$Q = \Phi^{-1}(1 - p_{fa}) \quad (6)$$

where Th is the local threshold for a local region, Q is a constant related to the setting false alarm probability p_{fa} , $\Phi^{-1}(\bullet)$ is the inverse function of standard normal distribution function. $\hat{\mu}_c$ and $\hat{\sigma}_c$ are the estimates of the local clutter mean and standard deviation,

$$\hat{\mu}_c = \frac{1}{N_c} \sum_{i,j \in \Omega_c} x(i,j) \quad (7)$$

$$\hat{\sigma}_c = \sqrt{\frac{1}{N_c} \sum_{i,j \in \Omega_c} (x(i,j) - \hat{\mu}_c)^2} \quad (8)$$

where $x(i,j)$ is the amplitude of pixel (i,j) . As shown in Fig 4, Ω_c represents the clutter region used to compute $\hat{\mu}_c$ and $\hat{\sigma}_c$. N_c is the number of pixels in Ω_c . If the amplitude mean ME of the local region exceeds the local threshold Th calculated in (5), the local region is considered to be a target region and preserved; if not, it is declared to be a clutter region and removed.

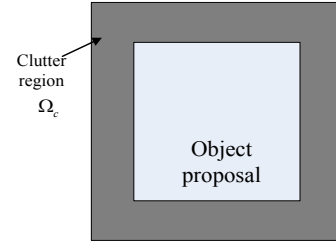


Fig. 4. The local regions based CFAR detector

III. EXPERIMENTS

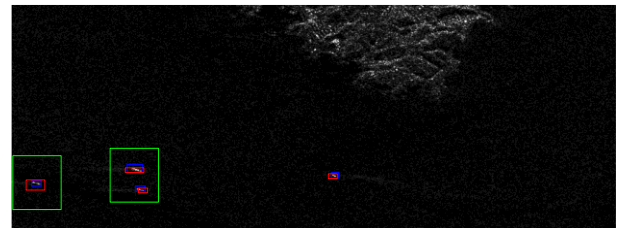
A. Experimental Images

In this section, we demonstrate the performance of our proposed ship targets detection method in SAR image on RADARSAT-2 SAR data. The RADARSAT-2 data used here were purchased from Center for Earth Observation and Digital Earth, Chinese Academy of Sciences. The two RADARSAT-2 SAR images used in the following experiments. The sizes of the two images are 4097×4297 and 1333×3687 , respectively. In the experiment, we cut these large images into twenty eight small images. Eight randomly selected images in which the sizes of ship targets vary greatly are used as training data, and the remaining twenty small images and the third large image are used as test images.

B. Detection Results

We evaluate the performance of our proposed method by comparison it with the traditional two-parameter CFAR. For the fair comparison, the traditional two-parameter CFAR is under the Gaussian distribution assumption and use the same clustering method in [2] to obtain the target regions in the following experiment.

In both the proposed method and classical two-parameter CFAR method, pfa is set as 10^{-6} . Fig. 5 gives the detection results on five test images by our proposed method and the classical two-parameter CFAR method, where the blue rectangles represent the ground truths, the green rectangles and red rectangles represent the detection results of the classical two-parameter CFAR method and our proposed method, respectively. Table I gives the detection results with our method and the classical two-parameter CFAR method on test images in Fig. 5. From Fig.5, we can see that there is no missing target in our proposed method, besides there is only one target in every detection window. However, the classical two-parameter CFAR method have false negatives in some test images. In addition, there are two or more targets in one detected target window when these targets locate closely via the classical two-parameter CFAR method.



(a)

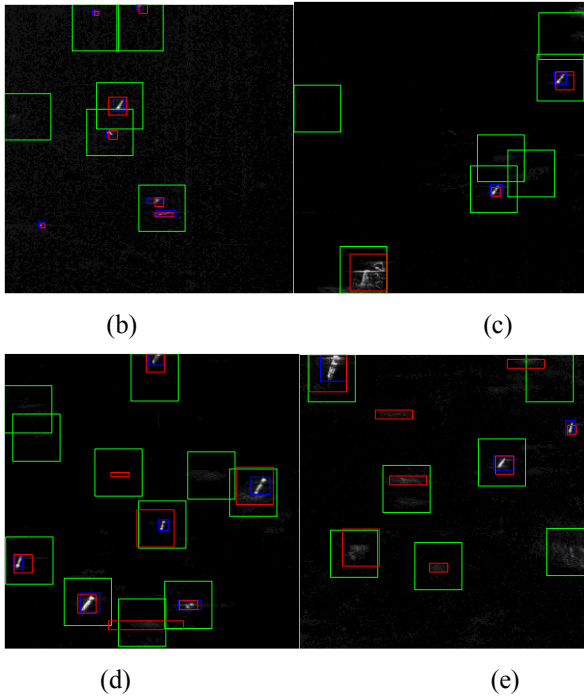


Fig. 5. Ship target detection results on the test images. The blue rectangles are ground truths. The green rectangles and the red rectangles are extracted by the traditional two-parameter CFAR and our proposed method, respectively.

From Table I, we can see that our proposed method has no false negatives. Therefore, the local region generation in our method achieves one hundred percent recall rate through all local regions on each test image. In addition, our proposed method has less false positives (FPs) than the classical two-parameter CFAR method for the test images in Fig. 5(b)(c)(d).

TABLE I

DETECTION PERFORMANCE COMPARISON BETWEEN OUR METHOD AND THE CLASSICAL TWO-PARAMETER CFAR METHOD ON TEST IMAGES IN FIG. 5

Test images in Fig. 5		(a)	(b)	(c)	(d)	(e)
Ground truth: Number of ship targets (NP)		4	7	2	6	3
Our proposed method	Number of detected ship targets	4	7	2	6	3
	Number of true positive windows (TP)	4	7	2	6	3
	Number of false positive windows (FP)	0	0	1	2	5
Two-parameter CFAR	Number of detected ship targets	3	6	2	6	2
	Number of true positive windows (TP)	2	5	2	6	2
	Number of false positive windows (FP)	0	1	5	5	5

IV. CONCLUSION

In this paper, we propose a ship target detection method for SAR image based on local region under the multi-scale and multi-target condition. Firstly, we use a local region generator based on NG feature to generate a small set of local regions. Then a local region based CFAR detector is

used to detect the true positive local regions. Finally, the final detection result is obtained by NMS. Experimental results on real SAR images demonstrate that our proposed method can get better performance than the classical two-parameter CFAR method in the multi-scale situation for ship target detection.

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